

Chapter 3

INDUCTION AND RECURSION

EXERCISE 3.1

Short Questions

Q.1 Solve / write answers of the following short questions:

(i) Use Mathematical induction to show that

$$1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$$

where n is non-negative integer.

PU, 2011: 2012 (BS Math)

Sol: $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1 \quad \dots(1)$

For $n = 0$, we have

$$1 = 2^{0+1} - 1 = 2 - 1 = 1$$

This shows that (1) is true for $n = 1$, so Condition-I (C-I) is satisfied.

Let (1) be true for $n = k \in W$, then

$$1 + 2 + 2^2 + \dots + 2^k = 2^{k+1} - 1 \quad \dots(2)$$

Adding 2^{k+1} on both sides, we have

$$\begin{aligned} 1 + 2 + 2^2 + \dots + 2^k + 2^{k+1} &= 2^{k+1} - 1 + 2^{k+1} \\ &= 2 \cdot 2^{k+1} - 1 \\ &= 2^{k+1+1} - 1 \end{aligned}$$

This shows that (1) is true for $n = k + 1$, so Condition-II (C-II) is satisfied. Consequently (1) is true for all non-negative integers n .

(ii) Show that $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$ for all positive integers n .

Sol: $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2} \quad \dots(1)$

For $n = 1$, we have

$$1 = \frac{1(1+1)}{2} = \frac{2}{2} = 1$$

This shows that (1) is true for $n = 1$, so Condition-I (C-I) is satisfied.

Let (1) be true for $n = k \in \mathbb{N}$, then

$$1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2} \quad \dots(2)$$

Adding $(k+1)$ on both sides, we have

$$\begin{aligned} 1 + 2 + 3 + \dots + k(k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2} \\ &= \frac{(k+1)(k+1+1)}{2} \end{aligned}$$

This shows that (1) is true for $n = k+1$, so Condition-II (C-II) is satisfied. Consequently (1) is true for all positive integers n .

(III) Show that $5 + 7 + 9 + \dots + (2n+3) = n(n+4)$ for all positive integers n .

Sol: $5 + 7 + 9 + \dots + (2n+3) = n(n+4) \quad (1)$

For $n = 1$, we have

$$5 = 1(1+4) = 5$$

This shows that (1) is true for $n = 1$, so Condition-I (C-I) is satisfied.

Let (1) be true for $n = k \in \mathbb{N}$, then

$$5 + 7 + 9 + \dots + (2k+3) = k(k+4) \quad (2)$$

Adding $[2(k+1)+3]$ on both sides, we have

$$\begin{aligned} 5 + 7 + 9 + \dots + (2k+3) + [2(k+1)+3] &= k(k+4) + [2(k+1)+3] \\ &= k(k+4) + 2k + 2 + 3 \\ &= k^2 + 4k + 2k + 5 \\ &= k^2 + 6k + 5 \\ &= k^2 + 5k + k + 5 \\ &= k(k+5) + 1(k+5) \\ &= (k+1)(k+5) \\ &= (k+1)(k+1+4) \end{aligned}$$

This shows that (1) is true for $n = k+1$, so Condition-II (C-II) is satisfied. Consequently (1) is true for all positive integers n .

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- (iv) Use Mathematical induction to show that
 $3 + 7 + 11 + \dots + (4n - 1) = n(2n + 1)$
 where n is a positive integer.

Sol: $3 + 7 + 11 + \dots + (4n - 1) = n(2n + 1)$ (1)

For $n = 1$, we have

$$3 = 1(2(1) + 1) = 2 + 1 = 3$$

This shows that (1) is true for $n = 1$, so Condition-I (C-I) is satisfied.

Let (1) be true for $n = k \in N$, then

$$3 + 7 + 11 + \dots + (4k - 1) = k(2k + 1)$$
 (2)

Adding $[4(k + 1) - 1]$ on both sides, we have

$$\begin{aligned} 3 + 7 + 11 + \dots + (4k - 1) + [4(k + 1) - 1] &= k(2k + 1) + [4(k + 1) - 1] \\ &= 2k^2 + k + 4k + 4 - 1 \\ &= 2k^2 + 5k + 3 \\ &= 2k^2 + 3k + 2k + 3 \\ &= k(2k + 3) + 1(2k + 3) \\ &= (k + 1)(2k + 3) \\ &= (k + 1)(2k + 2 + 1) \\ &= (k + 1)[2(k + 1) + 1] \end{aligned}$$

This shows that (1) is true for $n = k + 1$, so Condition-II (C-II) is satisfied. Consequently (1) is true for all positive integers n .

- (v) Let $P(n)$ be the statement that

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n + 1)(2n + 1)}{6}$$

for the positive integer n .

What is the statement $P(1)$?

Sol: The statement $P(1)$ is

$$1^2 = \frac{1(1 + 1)(2 + 1)}{6}$$

- (vi) Let $P(n)$ be the statement that

$$1.2 + 2.3 + \dots + n(n + 1) = \frac{n(n + 1)(n + 2)}{3}$$

for the positive integer n .

What is the statement $P(2)$?

Sol: The statement $P(2)$ is

$$1.2 + 2.3 = \frac{2(2 + 1)(2 + 2)}{3}$$

- (vii) Prove that $1.1! + 2.2! + \dots + n.n! = (n + 1)! - 1$, whenever n is a positive integer.

Sol: $1 \cdot 1 + 2 \cdot 2 + \dots + n \cdot n! = (n+1)! - 1$... (1)

For $n = 1$, we have

$$1 \cdot 1 = (1+1)! - 1$$

$$1 = 2! - 1 = 2 - 1 = 1$$

This shows that (1) is true for $n = 1$, so Condition-I (C-I) is satisfied.

Let (1) be true for $n = k \in N$, then

$$1 \cdot 1 + 2 \cdot 2 + \dots + k \cdot k! = (k+1)! - 1$$
 ... (2)

Adding $(k+1) \cdot (k+1)!$ on both sides, we have

$$1 \cdot 1 + 2 \cdot 2 + \dots + k \cdot k! + (k+1) \cdot (k+1)! = (k+1)! - 1 + (k+1) \cdot (k+1)!$$

$$= (k+1)! + (k+1) \cdot (k+1)! - 1$$

$$= (k+1)! [1 + (k+1)] - 1$$

$$= (k+1)! (k+1+1) - 1$$

$$= (k+1+1)! - 1$$

This shows that (1) is true for $n = k+1$, so Condition-II (C-II) is satisfied. Consequently (1) is true for all positive integers n .

(viii) **Prove that 2 divides $n^2 + n$ whenever n is a positive integer.**

Sol: For $n = 1$, we have

$$\frac{n^2 + n}{2} = \frac{1^2 + 1}{2} = \frac{2}{2}$$

This shows that 2 divides $n^2 + n$ for $n = 1$, so C-I is satisfied.

Let 2 divide $n^2 + n$ for $n = k \in N$, then

$$\frac{k^2 + k}{2} = \text{Integer}$$
 (1)

Next consider

$$\frac{(k+1)^2 + (k+1)}{2} = \frac{k^2 + 2k + 1 + k + 1}{2}$$

$$= \frac{k^2 + k + 2(k+1)}{2}$$

$$= \frac{k^2 + k}{2} + \frac{2(k+1)}{2}$$

$$= \text{Integer} + (k+1) \quad (\text{using (1)})$$

$$= \text{Integer}$$

This shows that 2 divides $n^2 + n$ for $n = k+1$, so C-II is satisfied.

Consequently 2 divides $n^2 + n$ for positive integers n .

(ix) **Prove that 3 divides $n^3 + 2n$ whenever n is a positive integer.**

Sol: For $n = 1$, we have $\frac{n^3 + 2n}{3} = \frac{1^3 + 2(1)}{3} = \frac{3}{3}$

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This shows that 3 divides $n^3 + 2n$ for $n = 1$, so C-I is satisfied.
 Let 3 divide $n^3 + 2n$ for $n = k \in N$, then

$$\frac{k^3 + 2k}{3} = \text{Integer} \quad (1)$$

Next consider

$$\begin{aligned} \frac{(k+1)^3 + 2(k+1)}{3} &= \frac{k^3 + 3k^2 + 3k + 1 + 2k + 2}{3} \\ &= \frac{k^3 + 2k + 3k^2 + 3k + 3}{3} \\ &= \frac{k^3 + 2k + 3(k^2 + k + 1)}{3} \\ &= \frac{k^3 + 2k}{3} + \frac{3(k^2 + k + 1)}{3} \\ &= \frac{k^3 + 2k}{3} + (k^2 + k + 1) \\ &= \text{Integer} + (k^2 + k + 1) \quad (\text{using (1)}) \\ &= \text{Integer} \end{aligned}$$

This shows that 3 divides $n^3 + 2n$ for $n = k + 1$, so C-II is satisfied.
 Consequently 3 divides $n^3 + 2n$ for positive integers n .

- (x) **Prove that 6 divides $n^3 - n$ whenever n is a nonnegative integer.**

Sol: For $n = 1$, we have

$$\frac{n^3 - n}{6} = \frac{1^3 - 1}{6} = \frac{0}{6}$$

This shows that 6 divides $n^3 - n$ for $n = 1$, so C-I is satisfied.
 Let 6 divide $n^3 - n$ for $n = k \in N$; then

$$\frac{k^3 - k}{6} = \text{Integer} \quad (1)$$

Next consider

$$\begin{aligned} \frac{(k+1)^3 - (k+1)}{6} &= \frac{k^3 + 3K^2 + 3K + 1 - k - 1}{6} \\ &= \frac{k^3 - k + 3K^2 + 3K}{6} \\ &= \frac{k^3 - k}{6} + \frac{3K^2 + 3K}{6} \\ &= \text{Integer} + \frac{3K(k+1)}{6} \quad (\text{using (1)}) \end{aligned}$$

$$\frac{(k+1)^3 - (k+1)}{6} = \text{Integer} + \frac{K(k+1)}{2}$$

$$\frac{(k+1)^3 - (k+1)}{6} = \text{Integer} + \text{Integer} \quad \therefore \frac{K(k+1)}{2} = \text{Integer}$$

$$= \text{Integer}$$

This shows that 6 divides $n^3 - n$ for $n = k+1$, so C-II is satisfied.
Consequently 6 divides $n^3 - n$ for positive integers n .

Long Questions

Q.2 Prove that for every positive integer n

$$1.2 + 2.3 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3}$$

PU, 2011 (BS Math)

Sol: $1.2 + 2.3 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3} \quad (1)$

For $n = 1$, we have

$$1.2 = \frac{1(1+1)(1+2)}{3}$$

$$2 = 2$$

This shows that (1) is true for $n = 1$, so C-I is satisfied.
Let (1) be true for $n = k \in \mathbb{N}$, then

$$1.2 + 2.3 + \dots + k(k+1) = \frac{k(k+1)(k+2)}{3} \quad (2)$$

Adding $(k+1)(k+2)$ on both sides, we have

$$\begin{aligned} 1.2 + 2.3 + \dots + k(k+1) + (k+1)(k+2) &= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2) \\ &= \frac{k(k+1)(k+2) + 3(k+1)(k+2)}{3} \\ &= \frac{(k+1)(k+2)(k+3)}{3} \\ &= \frac{(k+1)(k+1+1)(k+1+2)}{3} \end{aligned}$$

This shows that (1) is true for $n = k+1$, so C-II is satisfied.
Consequently (1) is true for all positive integers n .

Q.3 Use mathematical induction to show that 5 divides $n^5 - n$ whenever n is a non-negative integer.

PU, 2012 (BS Math)

Sol: For $n = 0$, we have $\frac{n^5 - n}{5} = \frac{0}{5} = 0$

This shows that 5 divides $n^5 - n$ for $n = 0$, so C-I is satisfied.

Let 5 divide $n^5 - n$ for $n = k \in W$, then

$$\frac{k^5 - k}{5} = \text{Integer} \quad (1)$$

Next consider

$$\begin{aligned} \frac{(k+1)^5 - (k+1)}{5} &= \frac{k^5 + 5K^4 + 10k^3 + 10k^2 + 5k + 1 - k - 1}{5} \\ &= \frac{k^5 - k + 5K^4 + 10k^3 + 10k^2 + 5k}{5} \\ &= \frac{k^5 - k}{5} + \frac{5K^4 + 10k^3 + 10k^2 + 5k}{5} \\ &= \text{Integer} + \frac{5k(K^3 + 2k^2 + 2k + 1)}{5} \quad (\text{using (1)}) \\ &= \text{Integer} + k(K^3 + 2k^2 + 2k + 1) \\ &= \text{Integer} + \text{Integer} \\ &= \text{Integer} \end{aligned}$$

This shows that 5 divides $n^5 - n$ for $n = k + 1$, so C-II is satisfied.

Consequently 5 divides $n^5 - n$ for positive integers n .

Q.4 Show that $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$ for all positive integers n .

PU. 2012 (BS IT)

Sol: $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6} \quad (1)$

For $n = 1$, we have

$$1^2 = \frac{1(1+1)(2(1)+1)}{6} = \frac{2(3)}{6} = 1$$

This shows that (1) is true for $n = 1$, so C-I is satisfied.

Let (1) be true for $n = k \in N$, then

$$1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6} \quad (2)$$

Adding $(k+1)^2$ on both sides, we have

$$\begin{aligned} 1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \\ &= \frac{k(k+1)(2k+1) + 6(k+1)^2}{6} \\ &= \frac{(k+1)[k(2k+1) + 6(k+1)]}{6} \end{aligned}$$

$$\begin{aligned}
1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 &= \frac{(k+1)(2k^2 + k + 6k + 6)}{6} \\
&= \frac{(k+1)(2k^2 + 7k + 6)}{6} \\
&= \frac{(k+1)(2k^2 + 4k + 3k + 6)}{6} \\
&= \frac{(k+1)[2k(k+2) + 3(k+2)]}{6} \\
&= \frac{(k+1)(k+2)(2k+3)}{6} \\
&= \frac{(k+1)(k+1+1)(2k+2+1)}{6} \\
&= \frac{(k+1)(\overline{k+1+1})(\overline{2k+1+1})}{6}
\end{aligned}$$

This shows that (1) is true for $n = k+1$, so C-II is satisfied. Consequently (1) is true for all positive integers n .

Q.5 Show that $1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$ for all positive integers n .

Sol: $1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2 \quad (1)$

For $n = 1$, we have

$$1^3 = \left[\frac{1(1+1)}{2} \right]^2 = 1$$

This shows that (1) is true for $n = 1$, so C-I is satisfied. Let (1) be true for $n = k \in N$, then

$$1^3 + 2^3 + 3^3 + \dots + k^3 = \left[\frac{k(k+1)}{2} \right]^2 \quad (2)$$

Adding $(k+1)^3$ on both sides, we have

$$\begin{aligned}
1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3 &= \left[\frac{k(k+1)}{2} \right]^2 + (k+1)^3 \\
&= \frac{k^2(k+1)^2}{4} + (k+1)^3 \\
&= \frac{k^2(k+1)^2 + 4(k+1)^3}{4}
\end{aligned}$$

$$\begin{aligned}
 1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3 &= \frac{(k+1)^2[k^2 + 4(k+1)]}{4} \\
 &= \frac{(k+1)^2[k^2 + 4k + 4]}{4} \\
 &= \frac{(k+1)^2(k+2)^2}{4} \\
 &= \frac{(k+1)^2(k+1+1)^2}{4} \\
 &= \left[\frac{(k+1)(k+1+1)}{2} \right]^2
 \end{aligned}$$

This shows that (1) is true for $n = k+1$, so C-II is satisfied. Consequently (1) is true for all positive integers n .

Q.6 Show that $1^2 + 3^2 + 5^2 + \dots + (2n+1)^2 = \frac{(n+1)(2n+1)(2n+3)}{3}$, where n is a nonnegative integer.

Sol: $1^2 + 3^2 + 5^2 + \dots + (2n+1)^2 = \frac{(n+1)(2n+1)(2n+3)}{3} \quad (1)$

For $n = 0$, we have

$$1^2 = \frac{(0+1)(0+1)(0+3)}{3} = 1$$

This shows that (1) is true for $n = 1$, so C-I is satisfied. Let (1) be true for $n = k \in W$, then

$$1^2 + 3^2 + 5^2 + \dots + (2k+1)^2 = \frac{(k+1)(2k+1)(2k+3)}{3} \quad (2)$$

Adding $[2(k+1)+1]^2$ on both sides, we have

$$\begin{aligned}
 1^2 + 3^2 + 5^2 + \dots + (2k+1)^2 + [2(k+1)+1]^2 &= \frac{(k+1)(2k+1)(2k+3)}{3} + [2(k+1)+1]^2 \\
 &= \frac{(k+1)(2k+1)(2k+3)}{3} + (2k+3)^2 \\
 &= \frac{(k+1)(2k+1)(2k+3) + 3(2k+3)^2}{3} \\
 &= \frac{[(k+1)(2k+1) + 3(2k+3)](2k+3)}{3} \\
 &= \frac{[2k^2 + 3k + 1 + 6k + 9](2k+3)}{3} \\
 &= \frac{[2k^2 + 9k + 10](2k+3)}{3}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{[2k^2 + 5k + 4k + 10](2k + 3)}{3} \\
 &= \frac{[k(2k + 5) + 2(2k + 5)](2k + 3)}{3} \\
 &= \frac{(k + 2)(2k + 5)(2k + 3)}{3} \\
 &= \frac{(k + 2)(2k + 3)(2k + 5)}{3} \\
 &= \frac{(k + 1 + 1)(2k + 2 + 1)(2k + 2 + 3)}{3} \\
 &= \frac{(\overline{k + 1} + 1)(\overline{2k + 1} + 1)(\overline{2k + 1} + 3)}{3}
 \end{aligned}$$

This shows that (1) is true for $n = k + 1$, so C-II is satisfied. Consequently (1) is true for all nonnegative integers n .

Q.7 Use mathematical induction to show that

$$5 + 10 + 20 + \dots + 5 \cdot 2^{n-1} = 5(2^n - 1)$$

for all positive integers n .

Sol: $5 + 10 + 20 + \dots + 5 \cdot 2^{n-1} = 5(2^n - 1)$ (1)

For $n = 1$, we have

$$5 = 5(2^1 - 1) = 5(2 - 1) = 5(1) = 5$$

This shows that (1) is true for $n = 1$, so C-I is satisfied.

Let (1) be true for $n = k \in \mathbb{N}$, then

$$5 + 10 + 20 + \dots + 5 \cdot 2^{k-1} = 5(2^k - 1)$$
 (2)

Adding $5 \cdot 2^k$ on both sides, we have

$$\begin{aligned}
 5 + 10 + 20 + \dots + 5 \cdot 2^{k-1} + 5 \cdot 2^k &= 5(2^k - 1) + 5 \cdot 2^k \\
 &= 5 \cdot 2^k - 5 + 5 \cdot 2^k \\
 &= 5 \cdot 2 \cdot 2^k - 5 \\
 &= 5 \cdot 2^{k+1} - 5 \\
 &= 5(2^{k+1} - 1)
 \end{aligned}$$

This shows that (1) is true for $n = k + 1$, so C-II is satisfied. Consequently (1) is true for all positive integers n .

Q.8 Use mathematical induction to show that

$$1 + 3 + 9 + \dots + 3^{n-1} = \frac{1}{2}(3^n - 1)$$

for all positive integers n .

Sol: $1 + 3 + 9 + \dots + 3^{n-1} = \frac{1}{2}(3^n - 1)$ (1)

For $n = 1$, we have

$$1 = \frac{1}{2}(3^1 - 1) = \frac{1}{2}(3 - 1) = \frac{1}{2}(2) = 1$$

This shows that (1) is true for $n = 1$, so C-I is satisfied.
Let (1) be true for $n = k \in \mathbb{N}$, then

$$1 + 3 + 9 + \dots + 3^{k-1} = \frac{1}{2}(3^k - 1) \quad (2)$$

Adding 3^k on both sides, we have

$$\begin{aligned} 1 + 3 + 9 + \dots + 3^{k-1} + 3^k &= \frac{1}{2}(3^k - 1) + 3^k \\ &= \frac{1}{2}[(3^k - 1) + 2 \cdot 3^k] \\ &= \frac{1}{2}(3 \cdot 3^k - 1) \\ &= \frac{1}{2}(3^{k+1} - 1) \end{aligned}$$

This shows that (1) is true for $n = k + 1$, so C-II is satisfied.
Consequently (1) is true for all positive integers n .

Q.9 Use mathematical induction to show that

$$1 + \frac{1}{3} + \frac{1}{9} + \dots + \frac{1}{3^{n-1}} = \frac{3}{2} \left[1 - \frac{1}{3^n} \right]$$

for all positive integers n .

Sol: $1 + \frac{1}{3} + \frac{1}{9} + \dots + \frac{1}{3^{n-1}} = \frac{3}{2} \left[1 - \frac{1}{3^n} \right] \quad (1)$

For $n = 1$, we have

$$1 = \frac{3}{2} \left[1 - \frac{1}{3^1} \right] = \frac{3}{2} \left[\frac{3-1}{3} \right] = \frac{3}{2} \left[\frac{2}{3} \right] = 1$$

This shows that (1) is true for $n = 1$, so C-I is satisfied.

Let (1) be true for $n = k \in \mathbb{N}$, then

$$1 + \frac{1}{3} + \frac{1}{9} + \dots + \frac{1}{3^{k-1}} = \frac{3}{2} \left[1 - \frac{1}{3^k} \right] \quad (2)$$

Adding $\frac{1}{3^k}$ on both sides, we have

$$\begin{aligned} 1 + \frac{1}{3} + \frac{1}{9} + \dots + \frac{1}{3^{k-1}} + \frac{1}{3^k} &= \frac{3}{2} \left[1 - \frac{1}{3^k} \right] + \frac{1}{3^k} \\ &= \frac{3}{2} \left[1 - \frac{1}{3^k} + \frac{2}{3 \cdot 3^k} \right] \\ &= \frac{3}{2} \left[1 - \frac{1}{3^k} + \frac{2}{3^{k+1}} \right] \end{aligned}$$

$$1 + \frac{1}{3} + \frac{1}{9} + \dots + \frac{1}{3^{k-1}} + \frac{1}{3^k} = \frac{3}{2} \left[1 - \frac{3-2}{3^{k+1}} \right]$$

$$= \frac{3}{2} \left[1 - \frac{1}{3^{k+1}} \right]$$

This shows that (1) is true for $n = k+1$, so C-II is satisfied. Consequently (1) is true for all positive integers n .

Q.10 Use mathematical induction to show that

$$\frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$

for all positive integers n .

Sol:
$$\frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1} \quad (1)$$

For $n = 1$, we have
$$\frac{1}{1.3} = \frac{1}{2(1)+1}$$

$$\frac{1}{3} = \frac{1}{3}$$

This shows that (1) is true for $n = 1$, so C-I is satisfied.

Let (1) be true for $n = k \in \mathbb{N}$, then

$$\frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots + \frac{1}{(2k-1)(2k+1)} = \frac{k}{2k+1} \quad (2)$$

Adding $\frac{1}{[2k+1-1][2k+1+1]}$ on both sides, we have

$$\begin{aligned} \frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots + \frac{1}{(2k-1)(2k+1)} + \frac{1}{[2k+1-1][2k+1+1]} \\ &= \frac{k}{2k+1} + \frac{1}{[2k+1-1][2k+1+1]} \\ &= \frac{k}{2k+1} + \frac{1}{(2k+2-1)(2k+2+1)} \\ &= \frac{k}{2k+1} + \frac{1}{(2k+1)(2k+3)} \\ &= \frac{k(2k+3)+1}{(2k+1)(2k+3)} = \frac{2k^2+3k+1}{(2k+1)(2k+3)} \\ &= \frac{2k^2+2k+k+1}{(2k+1)(2k+3)} = \frac{2k(k+1)+1(k+1)}{(2k+1)(2k+3)} \\ &= \frac{(2k+1)(k+1)}{(2k+1)(2k+3)} = \frac{k+1}{2k+1+1} \end{aligned}$$

This shows that (1) is true for $n = k+1$, so C-II is satisfied.

Consequently (1) is true for all positive integers n .

Q.11 Use mathematical induction to show that

$$\binom{2}{2} + \binom{3}{2} + \binom{4}{2} + \dots + \binom{n+1}{2} = \binom{n+2}{3}$$

for all positive integers n .

Sol:
$$\binom{2}{2} + \binom{3}{2} + \binom{4}{2} + \dots + \binom{n+1}{2} = \binom{n+2}{3} \quad (1)$$

For $n = 1$, we have

$$\binom{2}{2} = \binom{1+2}{3}$$

$$1 = 1$$

This shows that (1) is true for $n = 1$, so C-I is satisfied.

Let (1) be true for $n = k \in N$, then

$$\binom{2}{2} + \binom{3}{2} + \binom{4}{2} + \dots + \binom{k+1}{2} = \binom{k+2}{3} \quad (2)$$

Adding $\binom{k+1+1}{2}$ on both sides, we have

$$\begin{aligned} \binom{2}{2} + \binom{3}{2} + \binom{4}{2} + \dots + \binom{k+1}{2} + \binom{k+1+1}{2} &= \binom{k+2}{3} + \binom{k+1+1}{2} \\ &= \binom{k+2}{3} + \binom{k+2}{2} \\ &= \binom{k+3}{3} \\ &= \binom{k+1+2}{3} \end{aligned}$$

This shows that (1) is true for $n = k+1$, so C-II is satisfied.

Consequently (1) is true for all positive integers n .

Q.12 Use mathematical induction to show that

$$\binom{5}{5} + \binom{6}{5} + \binom{7}{5} + \dots + \binom{n+4}{5} = \binom{n+5}{6}$$

for all positive integers n .

Sol:
$$\binom{5}{5} + \binom{6}{5} + \binom{7}{5} + \dots + \binom{n+4}{5} = \binom{n+5}{6} \quad (1)$$

For $n = 1$, we have

$$\binom{5}{5} = \binom{1+5}{6}$$

$$1 = 1$$

This shows that (1) is true for $n = 1$, so C-I is satisfied.

Let (1) be true for $n = k \in N$, then

$$\binom{5}{5} + \binom{6}{5} + \binom{7}{5} + \dots + \binom{k+4}{5} = \binom{k+5}{6} \quad (2)$$

Adding $\binom{k+1+4}{5}$ on both sides, we have

$$\begin{aligned} \binom{5}{5} + \binom{6}{5} + \binom{7}{5} + \dots + \binom{k+4}{5} + \binom{k+1+4}{5} &= \binom{k+5}{6} + \binom{k+1+4}{5} \\ &= \binom{k+5}{6} + \binom{k+5}{5} \\ &= \binom{k+6}{6} \\ &= \binom{k+1+5}{6} \end{aligned}$$

This shows that (1) is true for $n = k+1$, so C-II is satisfied.

Consequently (1) is true for all positive integers n .

Q.13 If n is any positive integer then show that $x + y$ is a factor of $x^{2n-1} + y^{2n-1}$, $x \neq -y$.

Sol: For $n = 1$, we have

$$x^{2n-1} + y^{2n-1} = x^{2-1} + y^{2-1} = x + y$$

This shows that $x + y$ is a factor of $x^{2n-1} + y^{2n-1}$ for $n = 1$.

Let $x + y$ be a factor of $x^{2n-1} + y^{2n-1}$ for $n = k \in N$, then

$$x^{2k-1} + y^{2k-1} = (x + y)Q(x, y) \quad (1)$$

where $Q(x, y)$ is a quotient when $x^{2k-1} + y^{2k-1}$ is divided by $x + y$.

Next consider

$$\begin{aligned} x^{2(k+1)-1} + y^{2(k+1)-1} &= x^{2k+2-1} + y^{2k+2-1} \\ &= x^{2k-1+2} + y^{2k-1+2} \\ &= x^2 x^{2k-1} + y^2 y^{2k-1} \\ &= x^2 x^{2k-1} + x^2 y^{2k-1} - x^2 y^{2k-1} + y^2 y^{2k-1} \\ &= x^2 (x^{2k-1} + y^{2k-1}) - y^{2k-1} (x^2 - y^2) \\ &= (x + y)Q(x, y) - y^{2k-1} (x^2 - y^2) \quad (\text{using (1)}) \\ &= (x + y)Q(x, y) - y^{2k-1} (x + y)(x - y) \end{aligned}$$

$$x^{2(k+1)-1} + y^{2(k+1)-1} = (x+y)[Q(x,y) - y^{2k-1}(x-y)]$$

This shows that $x+y$ is a factor of $x^{2n-1} + y^{2n-1}$ for $n = k+1$, so C-II is satisfied. Consequently $x+y$ is a factor of $x^{2n-1} + y^{2n-1}$ for all positive integers n .

Q.14 Show that $\frac{5^{2n-1} + 3^{2n-1}}{8}$ is an integer for any positive integer n .

Sol: For $n = 1$, we have

$$\frac{5^{2n-1} + 3^{2n-1}}{8} = \frac{5^{2-1} + 3^{2-1}}{8} = \frac{5^1 + 3^1}{8} = \frac{8}{8} = 1$$

This shows that $\frac{5^{2n-1} + 3^{2n-1}}{8}$ is an integer for $n = 1$, so C-I is satisfied.

Let $\frac{5^{2n-1} + 3^{2n-1}}{8}$ be an integer for $n = k \in \mathbb{N}$, then

$$\frac{5^{2k-1} + 3^{2k-1}}{8} = \text{Integer} \quad (1)$$

Next consider

$$\begin{aligned} \frac{5^{2(k+1)-1} + 3^{2(k+1)-1}}{8} &= \frac{5^{2k+2-1} + 3^{2k+2-1}}{8} = \frac{5^2 5^{2k-1} + 3^2 3^{2k-1}}{8} \\ &= \frac{25 \cdot 5^{2k-1} + 9 \cdot 3^{2k-1}}{8} \\ &= \frac{5^{2k-1} + 24 \cdot 5^{2k-1} + 3^{2k-1} + 8 \cdot 3^{2k-1}}{8} \\ &= \frac{5^{2k-1} + 3^{2k-1} + 24 \cdot 5^{2k-1} + 8 \cdot 3^{2k-1}}{8} \\ &= \frac{5^{2k-1} + 3^{2k-1}}{8} + \frac{24 \cdot 5^{2k-1} + 8 \cdot 3^{2k-1}}{8} \\ &= \text{Integer} + \frac{8(3 \cdot 5^{2k-1} + 3^{2k-1})}{8} \quad (\text{using (1)}) \\ &= \text{Integer} + 3 \cdot 5^{2k-1} + 3^{2k-1} \\ &= \text{Integer} + \text{Integer} \because 3 \cdot 5^{2k-1} + 3^{2k-1} = \text{Integer} \\ &= \text{Integer} \because \text{Integer} + \text{Integer} = \text{Integer} \end{aligned}$$

This shows that $\frac{5^{2n-1} + 3^{2n-1}}{8}$ is an integer for $n = k+1$, so C-II is



















